

PAUL MITCHELL

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DEFENSE TECHNOLOGY · SECURE SYSTEMS & DISTRIBUTED INFRASTRUCTURE

PROFESSIONAL SUMMARY

William & Mary Computer Science M.S. candidate (3.95 GPA, Systems & AI/ML) and clearable U.S. citizen targeting defense intelligence technology and STEM programs. Combines hands-on Linux systems administration (100+ servers), configuration automation (Puppet), and cluster observability (Prometheus/Grafana) with pre-inference AI defense architectures and secure, low-latency data streams. Proven record of hardening mission infrastructure, eliminating security drift, and engineering resilient telemetry.

EDUCATION

College of William & Mary — M.S. in Computer Science (AI/ML & Systems) Expected Dec 2026 · GPA: 3.95
College of William & Mary — B.S. in Computer Science Graduated Dec 2024
Relevant Coursework: Machine Learning Systems, Advanced Software Engineering, Deep Learning, Distributed Systems, Data Mining

TECHNICAL SKILLS & INFRASTRUCTURE

Systems & Infrastructure: Linux Systems Administration (100+ nodes), Puppet, Bash, SSH, DNS, Docker, Kubernetes, Netplan, Git
Security & AI Defense: Pre-inference LLM proxy, zero-width steganography decoding, honeypot canaries, STIX 2.1, ArcSight CEF
Observability & Metrics: Prometheus, Grafana, OpenTelemetry, alert rule design, incident triage, log auditing, capacity planning
Programming & Data: Python, FastAPI, C/C++, Java, SQL (PostgreSQL), PyTorch, LightGBM, AWS (S3, EC2), YAML

TECHNICAL & PROFESSIONAL EXPERIENCE

IT Administrator | William & Mary Technology Support Center | Williamsburg, VA Feb 2025 – Present

- Administer and maintain 100+ Linux instructional compute servers, orchestrating automated system configurations via Puppet, DNS management, and network troubleshooting across multi-room labs.
- Architected Prometheus and Grafana telemetry pipelines monitoring cluster uptime, storage utilization, and compute health, cutting incident detection and response times by over 40%.
- Evaluated multi-user Docker containerization architectures, producing technical documentation on isolation boundaries, resource limits, and operational maintainability.
- Standardized network routing and static IP configurations, maintaining strict configuration control across all production instructional hosts.
- Authored automated host health probe daemons in Bash, proactively detecting dead lockups and filesystem corruption.

Data Analytics & Engineering Extern | Extern, Amazon-Sponsored Project | Remote Jun 2025 – Sep 2025

- Engineered reproducible automated Python and AWS S3 data pipelines to clean, validate, and structure ~20,000 text records, optimizing data prep execution cycles from hours to minutes.
- Developed a predictive classification model using LightGBM and deployed an interactive Streamlit analytics platform delivering real-time decision-support metrics for program mentors.
- Implemented cloud dataset versioning using DVC on AWS S3 to ensure reproducibility and audit compliance across multi-team evaluations.
- Fine-tuned DistilBERT transformer models ($F1 = 0.89$) to classify textual incident narratives into actionable risk categories.

KEY TECHNICAL & SYSTEMS PROJECTS

ShadowPrompt: Pre-Inference Defense Proxy for Enterprise LLMs | Python, FastAPI, Three.js 2026 · pjmitchell7.github.io/shadowprompt

- Architected an in-memory pre-inference proxy inspecting raw Unicode streams at 0.038ms P99 latency, blocking steganography, directional overrides, and template breakouts before tokenization.
- Integrated deceptive honeypot sandboxes seeded with canary credentials to waste adversary compute without triggering evasion routines; exported compliant STIX 2.1 and ArcSight CEF syslog feeds.

PromptShield: Tokenization-Aware LLM Security Middleware | Python, FastAPI, scikit-learn 2026

- Built asynchronous FastAPI middleware intercepting adversarial prompt injections with 0.27–1.11ms latency, avoiding expensive secondary LLM calls while holding false positives strictly under 0.1%.
- Calibrated classification thresholds against high-volume adversarial traffic to enforce strict zero-day prompt rejection.

ChaosLedger: Zero-Discrepancy Stream Engine under Simulated AWS Partitioning | Python, Kafka, Redis 2026

- Architected a distributed financial stream engine simulating 100k tx/min under chaos-injected 200ms latency spikes; guaranteed \$0.00 balance discrepancy using atomic Redis idempotency.
- Maintained persistent partition replication under simulated node failures, preventing stream data corruption.